Lesson Plan — Session 2 · Metric Conversions (liters → milliliters)

Mr. Neft · Grade 6 Mathematics · Next class session after Monday, October 5, 2026 [VERIFY DATE] · 90-minute block

Reveal Math: Ratio Reasoning: Convert Measurements within the Same System · Source: 3.6.pptx slides 22–39 · Practice/3.6 Session 2.docx · HW/3.6 Session 2 HW.docx

1. DO NOW

Do Now — Review from Session 1 (Slide 22) · 4 minutes	Talk stem (TWR)
In Riding the Attractions, we changed 4 feet into 48 inches to compare with the 44-inch sign. Use the same move. 1. 1 foot = 12 inches · 3 feet = _____ inches 2. 1 liter = 1,000 milliliters · 4 liters = _____ milliliters 3. 48 inches = _____ feet 4. 2,000 milliliters = _____ liters	*"In Riding the Attractions, I multiplied 4 feet by ___ to get ___ inches. For 48 inches to feet, I ___ by 12."* **Key:** 36 · 4,000 · 4 · 2

2. STANDARDS

Grade 5 Prerequisite	Grade 6 Target Standard	Grade 7 Extension
5.MD.A.1 — Convert among different-sized standard measurement units within a given measurement system (e.g., 5 cm to 0.05 m) and use these conversions in multi-step, real-world problems.	**Maryland 6.AT.3** (deck, slides 3 and 24) · **CCSS 6.RP.A.3d** (HW and Answer Key) — Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities. Deck target (slide 39): *I can use ratio reasoning to convert between units of the same measurement system.*	**7.RP.A.1** — Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units.

3. OBJECTIVES

Objective	Statement	Self-check
Content	I can **convert** liters to milliliters with the **conversion factor** 1 L = 1,000 mL on a double number line, then divide to solve a capacity problem.	☐ Before | ☐ After
Language (WIDA)	I will **explain** each step of a conversion, labeling liters and milliliters, using the frame *"I changed ___ to ___ so I could compare ___"* when I share with a partner.	☐ Before | ☐ After

4. LESSON PROCEDURES

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Warm-Up (8 min)	• Display the Session 2 Warm-Up (Slide 22). Students solve the 4 items solo; mark one unsure item. [T3] • Ask: "Which item uses the same move as Riding the Attractions?" (item 1: 3 × 12) • Carryover check: partners compare item 3 (48 inches = 4 feet) and defend multiply or divide before the answers are shown. [T4★]	• Solve items 1–4: 36 · 4,000 · 4 · 2. • Defend item 3: *"In Riding the Attractions I multiplied by ___, but 48 inches to feet goes to a ___ unit, so I ___."*	• R.M.G. — visual schedule: Session 2 agenda posted and pointed to. • J.L. — clear expectations: "4 problems · 4 minutes" timer. • A.M.M. — wait time before any call-on; item 3 checked privately.	• L1–2: Use the Session 1 ratio table strip to answer items 1 and 3. • L3–4: Use the item-3 frame. • L5: Explain why items 1 and 3 are inverse moves. Sentence Frame: "I know this because..."	• [T4] Scan items 3 and 4 (the dividing items) for every student. • [T3] "How many milliliters are in 1 liter?" (1,000)
Launch (7 min)	• Review Notes — Session 2 (Slide 23): cover one line and explain it from memory, using Riding the Attractions. [T4] • Learning Target — Session 2 (Slide 24): read the content and language objectives. [T3] • Revisit Measurement system (Slide 8). Introduce kilo- = 1,000, centi- = 1/100, milli- = 1/1,000 from the Practice sheet vocabulary box.	• Explain a covered note line: *"Line ___ says ___; in Riding the Attractions that was 4 feet = ___ inches."* • Connect the systems: *"Session 1 used customary units like ___; today we use metric units like ___."*	• J.C. — pre-teaching: prefix card (kilo-, centi-, milli-) previewed before class. • M.E. — word bank: liter, milliliter, metric, kilo-, milli-. • S.D.L.P. — verbal response for the covered-line explanation.	• L1–2: Match prefix cards to 1,000 · 1/100 · 1/1,000; repeat aloud. • L3–4: Use the systems frame. • L5: Explain why metric prefixes make conversions faster than customary ones. Sentence Frame: "I know this because..."	• [T4] Listen for "the amount stayed the same" in the covered-line explanations. • [T3] "What does milli- mean?" (one thousandth)
Be Curious (10 min)	• Be Curious (Slide 25): fish above a ruler. Quiet look; collect notices and wonders. [T3] • Launch Hydrating (Slide 26): Diamond wants 4 liters a day; her bottle holds 500 milliliters. Students plan before calculating. [T4★] • A/B: A shares a plan; B asks which unit comes first. Record plans without confirming.	• Notice and wonder: *"I notice the ruler is marked in ___. I wonder how long the fish is in ___."* • Plan (Slide 26 frame): *"First I will convert ___ to ___. Then I will find how many groups of ___ fit."*	• R.B. — simplified language card: "Goal: 4 L. Bottle: 500 mL. Same unit first." • J.L. — movement: hold and pass a real 500 mL bottle around the table. • A.M.M. — safe response: plans shared with a partner before the class.	• L1–2: Point to 4 liters and 500 milliliters; circle the unit to change. • L3–4: Use the Slide 26 plan frame. • L5: Explain why 4 ÷ 500 would give a wrong answer. Sentence Frame: "I know this because..."	• [T4] Count the plans that convert before dividing. • [T3] "Which two units are in the Hydrating problem?"

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Guided Practice (15 min)	• Reveal sequence (Slides 33–35): first step = find the milliliters in 4 liters; tool = double number line; read the 4-liter mark. [T3] • Liters → Milliliters (Slides 27–28): 1 L = 1,000 mL; multiply both by 4; 4,000 mL = 4 L. Name Vocabulary 6 (Slide 9). [T4] • Milliliters → Bottle Fills (Slide 29): 4,000 ÷ 500 = 8. A explains why we divide; B asks for evidence. [T4★] • Today's Math Notes — Session 2 (Slide 30): copy 3 lines; cover one and explain.	• Complete the note-sheet double number line: Liters 0–4 matched to Milliliters 0, 1,000, ___, ___, ___. • Explain (Slide 27 frame): *"Both quantities were multiplied by ___. So 4 liters equals ___ milliliters."* • Explain (Slide 29 frame): *"I divided ___ by ___ because each full bottle holds ___."*	• S.D.L.P. — manipulatives: 1-liter bottle and 500 mL measuring cup (two fills per liter). • R.B. — graphic organizer: pre-drawn double number line template. • M.E. — chunking: copy note line 01, then 02, then 03 with checks between.	• L1–2: Place number cards on the double number line; say "4 liters equals 4,000 milliliters." • L3–4: Use the Slide 27 and Slide 29 frames. • L5: Explain why the answer is 8 fills, not 4,000. Sentence Frame: "I know this because..."	• [T4] Check that every double number line pairs 4 with 4,000. • [T3] "Why do we divide by 500?"
Collaborative Practice (15 min)	• Let's Explore More (Slide 31): graph of liters vs. milliliters; connect a point to the double number line; compare metric factors. [T4★] • Assign Practice sheet items 2–4 (5K Charity Run, Science Lab Mass, Science Class Ribbon) A/B. • Check Your Work (Slide 32): label units at every step and ask whether the count should increase. [T4]	• Slide 31: *"The point (___, ___) matches the ratio ___ because ___."* and *"These metric conversions use powers of ten. The factor changes because ___."* • Items 2–4 (5,000 m, 1,800 m left · 350,000 mg, 0.35 kg · 240 cm, 8 pieces). Partner check: *"Meters are ___ than kilometers, so the number should get ___."*	• R.B. — graphic organizer: graph and double number line printed side by side. • R.M.G. — predictability: same A/B roles as in Session 1. • S.D.L.P. — calculator for 2.4 × 100 and 240 ÷ 30.	• L1–2: Plot (1, 1,000) and (4, 4,000) with a partner; say each point. • L3–4: Use the Slide 31 frames. • L5: Compare the factors 10, 100, and 1,000 in writing.	• [T4] Listen for partners connecting a graph point to a double-number-line pair. • [T3] "What number do you multiply by to change kilometers to meters?" (1,000)
Independent Practice (12 min)	• Launch the three stations in Section 7 (rotation chart posted). • Lead Reteach (Practice items 1, 10, 11) with the bottle and measuring cup. [T4] • At minute 6, check the Practice station's item 7 (4,800 g against the 5 kg limit).	• Work the assigned station set; label units on every step. • Before each item: *"___ are smaller than ___, so I ___ by ___ and shift the decimal ___ places."*	• M.E. — extended time on a core set (items 1, 10, 11). • A.M.M. — 1:1 check-in after the first item. • R.M.G. — visual station schedule and predictable rotation signal.	• L1–2: Items marked with a big-to-small arrow picture; answer with number and unit. • L3–4: Use the direction frame. • L5: Item 13 — explain Daniel's error in writing.	• [T4] Check items 6 and 9 (dividing by 1,000) at the Practice station. • [T3] "Is a milliliter bigger or smaller than a liter? So will 2.5 get bigger or smaller?"
Closing (8 min)	• Summarize: Convert Units (Slide 36): "How do you decide whether to multiply or divide when converting?" Each student teaches a partner using one Session 1 and one Session 2 example. [T4★] • Learning Targets (Slide 39): self-rate both targets. • Assign HW/3.6 Session 2 HW.	• Teach a partner: *"In Riding the Attractions I multiplied by ___; in Hydrating I multiplied by ___ and then divided by ___, because ___."* • Self-rate both targets 1–4.	• J.C. — verbal response for the teach-a-partner explanation. • J.L. — reinforcement: name a specific strong explanation aloud. • R.M.G. — predictability: preview Lesson 3-7 on the agenda.	• L1–2: Point to the two examples on Slide 36 and say "multiply" or "divide." • L3–4: Use the teach-a-partner frame. • L5: Generalize a rule for any same-system conversion.	• [T4] Listen for both examples, Session 1 and Session 2, in each partner explanation. • [T3] "What is the conversion factor for liters and milliliters?"

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Exit Ticket (15 min)	• Assign tiers (Section 9) from the Collaborative Practice evidence. • Silent and independent; read aloud on request. • Collect and sort for the Lesson 3-7 groups. [T4]	• Complete the assigned tier; label every quantity. • Slide 38 frame: *"I changed ___ to ___ so I could compare ___."*	• M.E. — extended time and word bank. • S.D.L.P. — calculator. • J.C. — text-to-speech or teacher read-aloud. • A.M.M. — 1:1 check-in and reassurance.	• L1–2: Tier 1 read aloud, with a double number line provided. • L3–4: Tier 2 with the frame. • L5: Tier 3.	• [T4] Score each ticket against the Section 9 key. • [T3] On collection: "Which unit did you change first?" • **Target: 80%+ convert 2 L = 2,000 mL and find 4 full bottles, with labeled units.**

Total: 90 minutes. TEACH codes: [T3] ask–respond–affirm · [T4] justify · [T4★] justify + peer discourse or written reasoning.

5. VOCABULARY

Tier	Term	Student-Friendly Definition	Morphology (root + affix)	EN-ES Cognate ✓/✗	Cross-Disciplinary Use
3	metric units	The base-10 units: meter, gram, liter.	metr (Greek, measure) + -ic	✓ unidades métricas	Science labs (grams, liters); sports (100-meter dash)
3	liter / milliliter	Metric units of volume; 1 liter = 1,000 milliliters.	milli- (thousandth) + liter	✓ litro / mililitro	Nutrition labels; science measurement
3	kilo- / centi- / milli-	Prefixes: kilo = 1,000 of them; centi = 1/100; milli = 1/1,000.	kilo (Greek, thousand); centi (Latin, hundred); milli (Latin, thousand)	✓ kilo- / centi- / mili-	Computers (kilobyte); money (cent); time (millisecond)
3	conversion factor	The pair of equal amounts you multiply or divide by, such as 1 L = 1,000 mL.	convert + -sion; fact (make) + -or	✓ factor de conversión	Science; currency exchange
3	double number line	Two number lines lined up so matching marks show the same amount in two units.	double (two-fold) + number + line	✓ recta numérica doble	Map scales; timelines in social studies
2	powers of ten	Numbers like 10, 100, 1,000 made by multiplying 10 by itself.	power (ability, strength) + ten	✗ (potencias de diez)	Science notation; place value

6. MISCONCEPTIONS + INTERVENTIONS

#	Student Belief	Cognitive Root	Teacher Corrective Move	Intervention Route
1	Carryover from Session 1: dividing when going to a smaller unit (Daniel: 3 m ÷ 100 = 0.03 cm).	Has the operation but not the unit-size check from Riding the Attractions.	Ask: "In Riding the Attractions, feet → inches made the number bigger. Meters → centimeters is the same direction. Bigger or smaller?"	Practice item 13 (Daniel's Work); Reteach station item 10 (7 km → m).
2	Dividing before converting: 4 ÷ 500 in Hydrating.	Divides across different units; no "same unit first" habit.	Return to Slide 33: "What could be the first step?" Convert to mL, then divide.	Slide 33 revisit; Reteach station item 1 (Water Cooler Bottling).
3	Shifting the decimal the wrong number of places (2.5 mL → 0.25 L).	Prefix meaning (milli = 1/1,000) not connected to the number of places.	Prefix card: milli = 1,000 per liter → 3 places. Model on a place-value chart.	Practice item 6 (Liquid Medicine Dose); HW item 8 (Kwame).

#	Student Belief	Cognitive Root	Teacher Corrective Move	Intervention Route
4	Reporting 4,000 mL as the answer to "how many fills."	Stops after the conversion and loses the question.	Underline the question on Slide 26; ask "What are we counting — milliliters or bottles?"	Slide 29 revisit; Practice item 4b (30 cm pieces).
5	Multiplying when going to a larger unit (Kim: 600 cm × 100 = 60,000 m).	Applies "metric = × power of ten" without checking direction.	Use the double number line: 600 cm sits far left of 100 m. Bigger unit → fewer.	HW item 12; Practice item 5 (2,450 m → km).

7. SMALL-GROUP STATIONS (RUN DURING INDEPENDENT PRACTICE)

Station	Who	Problems	Guiding Question / Protocol	Expected Artifact	TWR Frame
RETEACH (teacher-led)	M.E., R.B., J.C., R.M.G., J.L., S.D.L.P., A.M.M. (full accommodations) + students who placed in the Session 1 exit-ticket Reteach pile	Practice sheet item 1 (Water Cooler Bottling: 4 L → 4,000 mL → 8 bottles) · item 10 (Race Route 7 km → 7,000 m) · item 11 (Water Bottle 1,500 mL → 1.5 L)	"Same unit first — which unit is smaller, and how many of it fit in one of the bigger unit?"	Double number line for item 1 plus two direction-labeled conversions.	*"Liters are ___ than milliliters, so I ___ by 1,000. 4 liters = ___ milliliters, and ___ ÷ 500 = ___ bottles."*
PRACTICE (partner)	Partners (A/B), same roles as Session 1	Practice sheet items 5–9 and 12 (2,450 m → 2.45 km · 2.5 mL → 0.0025 L · 4,800 g = 4.8 kg, under 5 kg · 12.75 kg → 12,750 g · 18,600 m → 18.6 km · 1.2 m → 120 cm → 1,200 mm)	A solves; B checks the direction and the number of decimal places before the number. Switch every item.	Answers with units and B's check mark per item.	*"You should shift the decimal ___ places to the ___ because ___ are ___ than ___."*
ENRICHMENT (independent)	Independent, Tier 3 on the Session 1 exit ticket	Apply: Making Concrete (Slide 37) · Practice item 14 (Class Relay: 8 × 450 = 3,600 m = 3.6 km)	Making Concrete mixes customary units: why do you divide by 16 there, but multiply by 1,000 in Hydrating?	A written paragraph with the full solution: 4 × 250 = 1,000 oz = 62.5 lb → 2 bags (60 lb each) → 2 × 1 1/2 gal = 3 gal = 12 qt.	*"I needed ___ bags because ___ pounds is more than ___ pounds. Each bag needs ___ gallons, so the water is ___ quarts."*

8. STUDENT NOTE SHEET

Lesson 3-6 · Converting Within One System · Session 2 Notes Name _____ Date _____ Period ____

A · Review from Session 1

Riding the Attractions: 4 feet = _____ inches, so the child _____ ride.

48 inches = _____ feet (I _____ by 12).

B · Hydrating (Slides 26–29)

Goal: 4 liters. Bottle: 500 milliliters.

My plan: "First I will convert ___ to ___. Then I will find how many groups of ___ fit."

Liters	0	1	2	3	4
Milliliters	0	1,000	___	___	___

4 liters = _____ mL. _____ ÷ 500 = _____ bottle fills.

C · My 3 Lines (Slide 30)

01 1 liter = _____ milliliters.

02 Use the same _____ before comparing amounts.

03 Check that the answer _____ the situation.

D · Metric Prefixes (Practice vocabulary)

kilo- = _____ centi- = _____ milli- = _____

3 m = _____ cm (multiply / divide) 2,450 m = _____ km (multiply / divide)

E · Explain It (TWR)

"In Session 1 I multiplied by ___ to change feet to inches. Today I multiplied by ___ to change liters to milliliters. Both times the number got bigger because _____."

Goal tracker: I can convert metric units with powers of ten and use the result to solve a capacity problem.	☐ 1 — Need help	☐ 2 — Need a hint	☐ 3 — Got it	☐ 4 — Can teach it

9. EXIT TICKET

Tier	Task	Frame	Answer Key
Tier 1 · Identify + frame	Exit Ticket (Slide 38): A container holds 2 liters. A bottle holds 500 milliliters. How many full bottles fill the container? (1 L = 1,000 mL)	*"I changed ___ to ___ so I could compare ___."*	2 L = 2,000 mL; 2,000 ÷ 500 = 4 bottles.

Tier	Task	Frame	Answer Key
Tier 2 · Identify + explain (TWR 2–3 sentences)	Aquarium Water Volume (Practice item 23): A tank holds 15 liters of water. How many 750 mL jugs fill the empty tank? Show the conversion, then explain in 2–3 sentences.	*"First I changed ___ liters to ___ milliliters because ___. Then I divided ___ by ___ to find ___."*	15 × 1,000 = 15,000 mL; 15,000 ÷ 750 = 20 jugs.
Tier 3 · Multi-variable + contrastive + transfer	Making Concrete (Slide 37): Each of 4 posts needs 250 ounces of concrete mix. One bag = 60 lb of mix and needs 1 1/2 gallons of water (1 lb = 16 oz; 1 gal = 4 qt). (a) How many bags? (b) How many quarts of water? (c) Contrast: Why did you divide by 16 here but multiply by 1,000 in Hydrating? (d) Transfer: Why do metric conversions only ever use 10, 100, or 1,000?	*"In Making Concrete I ___ because ounces are ___ than pounds; in Hydrating I ___ because liters are ___ than milliliters."*	(a) 4 × 250 = 1,000 oz; 1,000 ÷ 16 = 62.5 lb > 60 lb, so 2 bags. (b) 2 × 1 1/2 = 3 gal = 12 qt. Also accept 6 1/4 qt if the student mixes only the 62.5 lb needed and justifies it. (c) Opposite directions: to a larger unit → divide; to a smaller unit → multiply. (d) Metric units are built on base-10 prefixes.

10. GRADE 6 IEP ACCOMMODATIONS MATRIX

Student	Full Accommodation List	Applied This Session (phase: support)	Small Group
M.E.	Sentence starters, word banks, chunking, extended time, manipulatives	• Launch: word bank: liter, milliliter, metric, kilo-, milli-. • Guided Practice: chunking: copy note line 01, then 02, then 03 with checks between. • Independent Practice: extended time on a core set (items 1, 10, 11). • Exit Ticket: extended time and word bank.	Reteach station (all 7 students)
R.B.	Graphic organizers, simplified language, small group, breaks	• Be Curious: simplified language card: "Goal: 4 L. Bottle: 500 mL. Same unit first." • Guided Practice: graphic organizer: pre-drawn double number line template. • Collaborative Practice: graphic organizer: graph and double number line printed side by side.	Reteach station (all 7 students)
J.C.	Text-to-speech, verbal responses, extended time, pre-teaching	• Launch: pre-teaching: prefix card (kilo-, centi-, milli-) previewed before class. • Closing: verbal response for the teach-a-partner explanation. • Exit Ticket: text-to-speech or teacher read-aloud.	Reteach station (all 7 students)
R.M.G.	Visual schedules, sensory mods, motor mods, predictability	• Warm-Up: visual schedule: Session 2 agenda posted and pointed to. • Collaborative Practice: predictability: same A/B roles as in Session 1. • Independent Practice: visual station schedule and predictable rotation signal. • Closing: predictability: preview Lesson 3-7 on the agenda.	Reteach station (all 7 students)
J.L.	Movement, clear expectations, reinforcement, chunking, visuals	• Warm-Up: clear expectations: "4 problems · 4 minutes" timer. • Be Curious: movement: hold and pass a real 500 mL bottle around the table. • Closing: reinforcement: name a specific strong explanation aloud.	Reteach station (all 7 students)
S.D.L.P.	Multi-sensory, manipulatives, verbal responses, calculator	• Launch: verbal response for the covered-line explanation. • Guided Practice: manipulatives: 1-liter bottle and 500 mL measuring cup (two fills per liter). • Collaborative Practice: calculator for 2.4 × 100 and 240 ÷ 30. • Exit Ticket: calculator.	Reteach station (all 7 students)
A.M.M.	Wait time, safe responses, 1:1 check-ins, reassurance	• Warm-Up: wait time before any call-on; item 3 checked privately. • Be Curious: safe response: plans shared with a partner before the class. • Independent Practice: 1:1 check-in after the first item. • Exit Ticket: 1:1 check-in and reassurance.	Reteach station (all 7 students)